Fundaments of Machine learning for and with engineering applications

Enrico Riccardi[1]

Department of Mathematics and Physics, University of Stavanger (UiS).[1]

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Course objectives

Understanding of

- data sources and consequent data properties.
- data analysis/machine learning approaches outcomes.
- sensitivity analysis
- predictive modeling
- multivariate data analysis
- machine learning techniques application
- ensemble methods
- Bayes approach
- visualization and reporting

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- Streaming tends to lead to empty classrooms and higher failure rate.

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- There will be a combination of lectures and tutorials.
- Tutorial and hands-on will be presented during the course.
- Study groups are strongly encouraged.
- In class discussions are encouraged at any stage.
- Flip classroom approach: problem first (when possible).
- Feedback expected from you!
- Note: each teacher, even if coordinated, will have a different approach/expectations.

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- You are **strongly advised** to construct your own Git repository (and to keep it in order).
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A personal consideration on generative AI

With great power comes great responsibility (Spider-Man)

LLMs can and shall be used. As their development is surging in the last years, their help in writing code and reports is undeniable. Students shall learn how to master these tools. Yet, while their usage is encouraged, the risk of excessively rely on them **DO** hinder learning.

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- Decisions are the hinge. What influences your decisions has value.

- Better results come from better decisions.

STATISTICAL argument

- Machine learning can be a useful ONLY when it helps making better decisions.

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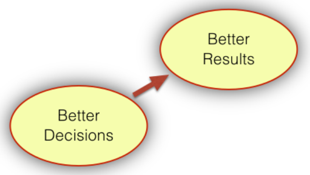
Life lesson?!

Life is a sum of all your choices (Albert Camus)

The only way you can purposefully influence your life, your family, your organization, your country or your world is through the decisions you make.



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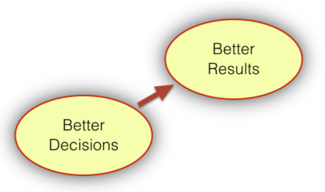

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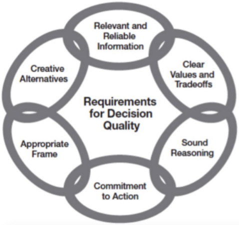

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- Please Do not use a technical approach for love matters.

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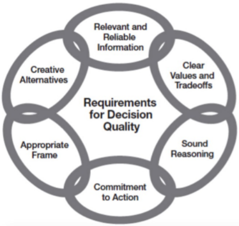

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You would thus need to:

1. Rationalize uncertainty (qualify)
2. Quantify uncertainty
3. Make decisions under uncertainty
4. Operate under uncertainty

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Uncertainty or Probability?

Our aim here is to provide good guidance on how to link data, models and output to value creation.

- First we need to understand uncertainty and probability, and the difference between the two.

- Spoiler: Probability is the language of uncertainty!

- We will analyse, quantify and structure models as a function of uncertainty.
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Data properties

- All starts from data: what are data-properties?

- Are there such things as good data and bad data?

Life lesson (or exam question, same thing ;))

- Data DO NOT always have value.

- TRASH in TRASH out

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- Relevance
- Significance
- Correlation
- Causation
- Data Filters
- Biases identification

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